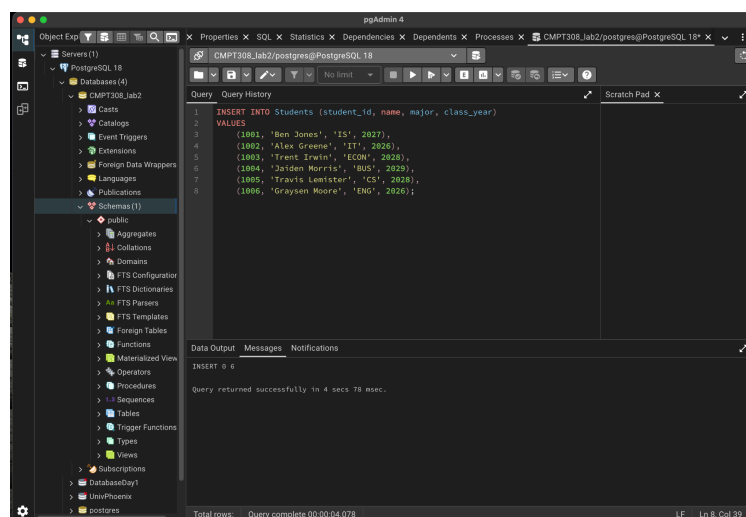
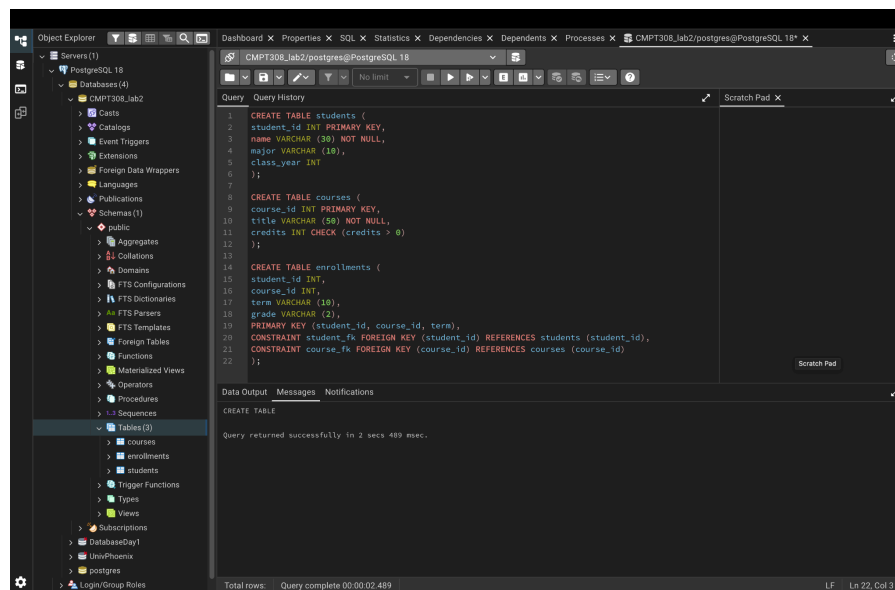
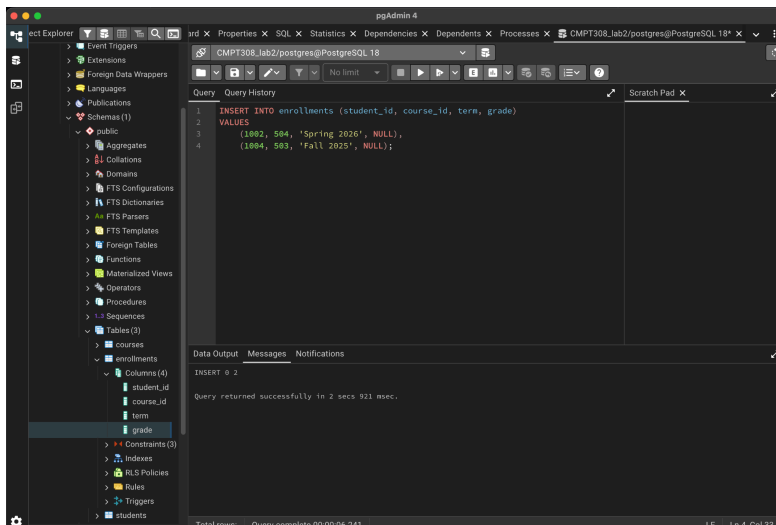
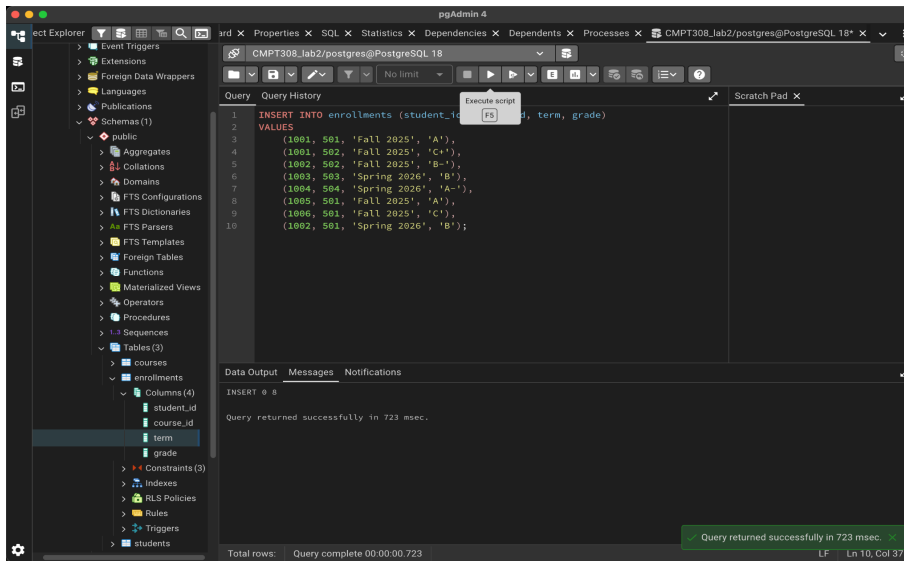
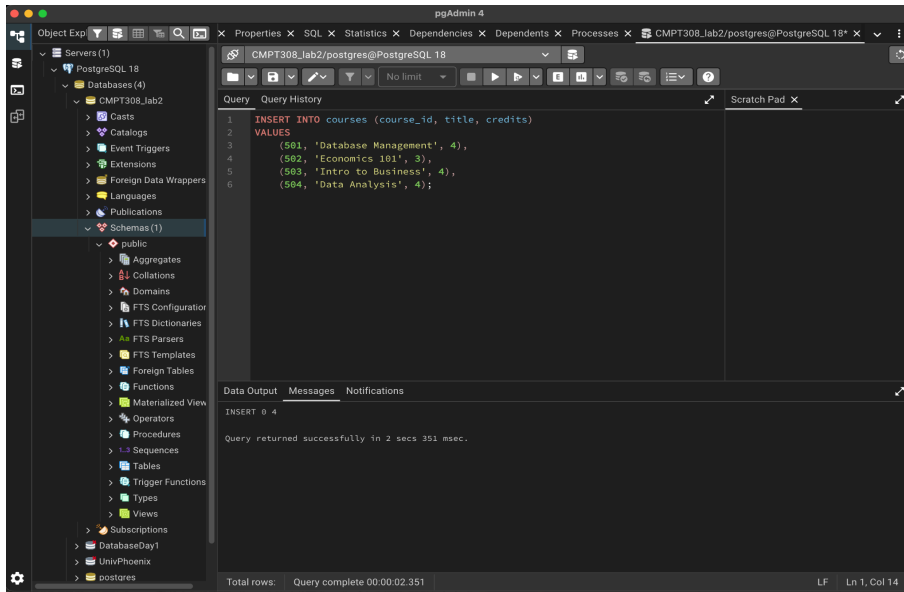


Part A:

1. A relation is a table, data set or a list.
2. A tuple is another name for a row that holds different attributes.
3. An attribute is what defines a column. Each column has its own unique attribute that helps make up the relation.
4. The domain is the allowed values for a given attribute. For example in this lab, student_id must be numeric, a positive number, length(4), and unique.
5. A schema refers to the design and structure of the table names, columns, and rows. An instance refers to the actual data in each tuple.
6. A key is an attribute or a set of attributes that uniquely identify a tuple. A primary key is a key from a single relation that uniquely distinguishes each tuple.

Part C:





pgAdmin 4

CMPT308_lab2/postgres@PostgreSQL 18*

Query: `SELECT * FROM courses`

Data Output: Messages Notifications

course_id [PK] integer	title character varying (50)	credits integer
1	501 Database Management	4
2	502 Economics 101	3
3	503 Intro to Business	4
4	504 Data Analysis	4

Total rows: 4 Query complete 00:00:13.238

pgAdmin 4

CMPT308_lab2/postgres@PostgreSQL 18*

Query: `SELECT * FROM students`

Data Output: Messages Notifications

student_id [PK] integer	name character varying (30)	major character varying (10)	class_year integer
1	1001 Ben Jones	IS	2027
2	1002 Alex Greene	IT	2026
3	1003 Trent Irwin	ECON	2028
4	1004 Jaden Morris	BUS	2029
5	1005 Travis Lemister	CS	2028
6	1006 Grayson Moore	ENG	2026

Total rows: 6 Query complete 00:00:03.314

pgAdmin 4

CMPT308_lab2/postgres@PostgreSQL 18*

Query: `SELECT * FROM enrollments`

Data Output: Messages Notifications

student_id [PK] integer	course_id [PK] integer	term [PK] character varying (20)	grade character varying (2)
1	1001	501 Fall 2025	A
2	1001	502 Fall 2025	C+
3	1002	502 Fall 2025	B-
4	1003	503 Spring 2026	B
5	1004	504 Spring 2026	A-
6	1005	501 Fall 2025	A
7	1006	501 Fall 2025	C
8	1002	501 Spring 2026	B
9	1002	504 Spring 2026	[null]
10	1004	503 Fall 2025	[null]

Total rows: 10 Query complete 00:00:05.866

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Query

Query History

1 SELECT * FROM students WHERE major = 'ECON';

Data Output

Messages

Notifications

Showing rows: 1 to 1

Page No:

	student_id [PK] integer	name character varying (30)	major character varying (10)	class_year integer
1	1003	Trent Irwin	ECON	2028

Query

Query History

1 SELECT student_id, name

2 FROM students

Data Output

Messages

Notifications

Showing rows: 1 to 1

Page No:

	student_id [PK] integer	name character varying (30)
1	1001	Ben Jones
2	1002	Alex Greene
3	1003	Trent Irwin
4	1004	Jaiden Morris
5	1005	Travis Lemister
6	1006	Graysen Moore

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Query

Query History

1 SELECT * FROM students WHERE name LIKE '%T%';

Data Output

Messages

Notifications

Showing rows: 1 to 2

Page No:

	student_id [PK] integer	name character varying (30)	major character varying (10)	class_year integer
1	1003	Trent Irwin	ECON	2028
2	1005	Travis Lemister	CS	2028

Query

Query History

1 SELECT * FROM enrollments WHERE grade IS NULL;

Data Output

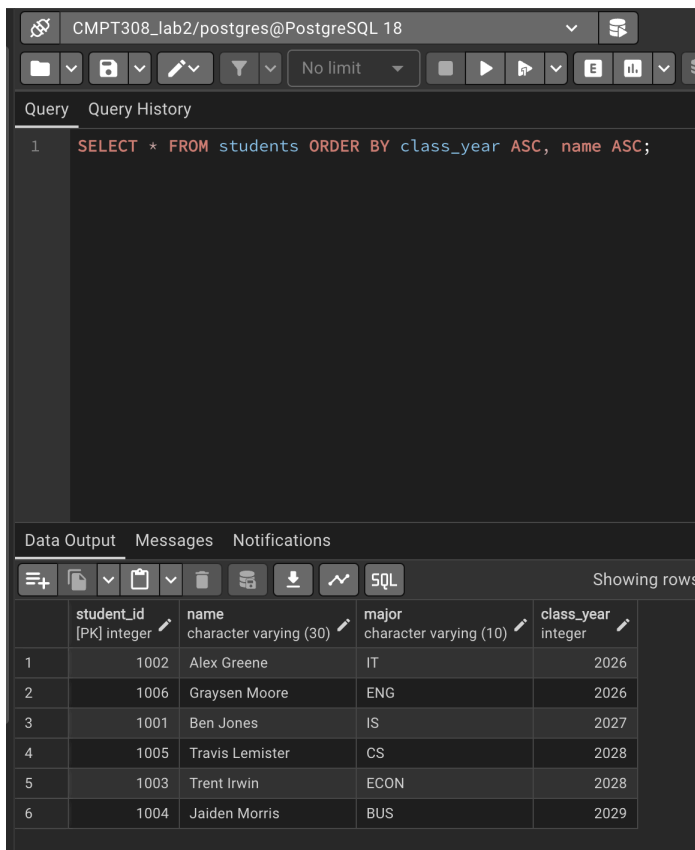
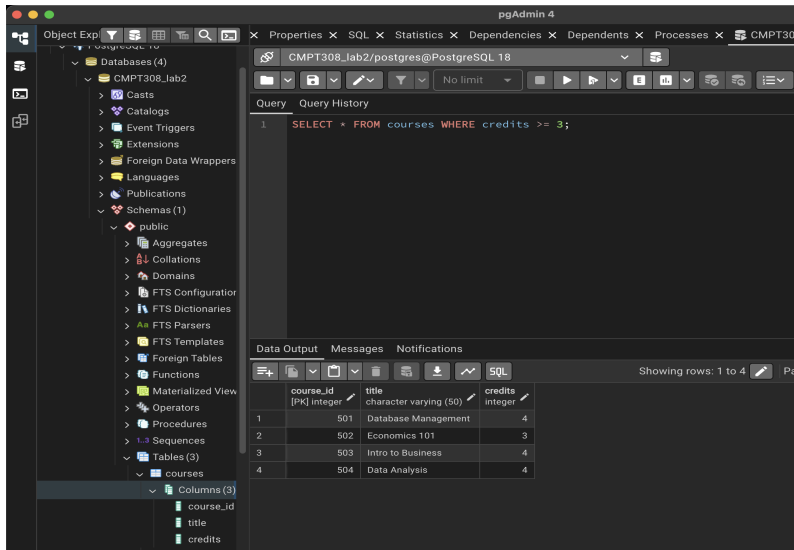
Messages

Notifications

Showing rows: 1 to 2

Page No:

	student_id [PK] integer	course_id [PK] integer	term [PK] character varying (20)	grade character varying (2)
1	1002	504	Spring 2026	[null]
2	1004	503	Fall 2025	[null]



Part D:

Select students whose major is a value you used.

$\sigma_{\text{major} = \text{'Econ'}}(\text{Students})$

List only student-id and name from (Students)

$\pi_{\text{student-id}, \text{name}}(\text{Students})$

https://github.com/TravisLemister/Database_Management_labs/blob/main/CMPT308_lab2